MAXIS: A Hyper-Efficient Paradigm for Scalable Long-Context LLM Training

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Abstract

Training Large Language Models (LLMs) with expansive vocabularies, often exceeding 256k tokens, presents a fundamental computational and memory bottleneck at the output projection layer. While Sampled Softmax approximations alleviate this overhead, they invariably succumb to "premature saturation," where the truncation of the partition function leads to vanishing gradients and stagnant convergence. We introduce **MAXIS** (Matryoshka Accelerated X-entropy Inference-ready Sampling), a novel training paradigm that recovers the rigorous mathematical supervision of full-vocabulary Cross-Entropy at a fraction of the compute cost. MAXIS replaces naive sampling with *Dynamic Variance Ghost Logits*, a learned probability sink that simulates the missing mass of the unsampled vocabulary in real-time, preventing gradient saturation and ensuring high-fidelity convergence. Furthermore, MAXIS imposes a hierarchical Matryoshka latent structure, natively optimizing the model for coarse-to-fine vector retrieval via Retrieval-Augmented Generation (RAG) by compressing knowledge into a 64-dimensional bottleneck. Empirical evaluations demonstrate that MAXIS achieves near-lossless convergence compared to state-of-the-art fused Cross-Entropy kernels, while delivering a 17.5× increase in throughput and a 38.7% reduction in peak VRAM, providing a universal framework for scalable, RAG-optimized LLM development.

1 Introduction

Modern Large Language Models (LLMs) based on the Transformer architecture [Vaswani et al., 2017] are governed by scaling laws [Hoffmann et al., 2022] that demand increasingly larger datasets and model capacities. A significant yet often overlooked scalability constraint is the final linear projection layer. Calculating Cross-Entropy (CE) over an expansive vocabulary V requires computing logits for every token in the sequence length T, a process that scales linearly with V. As models scale to support diverse languages and specialized domains, the resulting memory and compute footprint often forces practitioners to accept suboptimal batch sizes or restrictive context lengths.

Sampled Softmax techniques provide a path to efficiency by evaluating the target token against only a small subset of K negative candidates. However, current implementations fail to achieve the convergence of exact CE. By artificially truncating the denominator of the Softmax function, these methods allow the model to trivially push the target token's probability P_{target} toward 1.0. Once $P_{target} \approx 1$, the gradient $(P_{target} - 1)$ effectively vanishes; the model ceases learning, having failed to accurately differentiate the target from the tens of thousands of other tokens in the vocabulary.

In this work, we present **MAXIS**, a hyper-efficient training paradigm that resolves the partition function truncation problem. We introduce the **Unnormalized Ghost Logit**, a dynamic scalar appended to the sampled logits that mathematically estimates the expected mass of the missing vocabulary distribution. Combined with a strided negative mining strategy and Matryoshka Representation Learning (MRL) [Kusupati et al., 2022], MAXIS maintains high-pressure gradient signals throughout the training process.

Beyond its training efficiency, MAXIS natively constructs hierarchical embedding spaces. By optimizing the hidden state h against both full-dimensional and 64-dimensional latent projections of the embedding weights W, we ensure the model organizes information such that the initial dimensions contain the most semantic signal. This produces "Inference-Ready" embeddings that enable faster vector search and native RAG capabilities [Lewis et al., 2020] directly from the transformer's latent states, establishing MaximusLLM as a unified solution for efficient LLM training and retrieval.

2 Related Work

Reducing the computational burden of the final softmax layer has been a long-standing challenge. Early approaches focused on static structural approximations, such as Hierarchical Softmax or Adaptive Softmax [Grave et al., 2017], which partition the vocabulary based on word frequency. While these methods reduce compute time during inference, they significantly complicate the training pipeline and often degrade the representations of rare tokens.

An alternative branch of research utilizes Sampled Softmax [Jean et al., 2014], which builds upon the principles of Noise Contrastive Estimation [Gutmann and Hyvärinen, 2010] to estimate the full partition function using a small subset of negative candidates. While highly efficient for large vocabularies, standard sampling suffers from gradient saturation, leading to suboptimal convergence compared to exact objectives. Recent hardware-aware approaches, such as the Liger Kernel [Engineering, 2024], focus on fusing the Cross-Entropy operations directly in CUDA to avoid memory materialization. While effective, fused kernels do not alter the underlying $O(V)$ mathematical complexity.

Finally, structuring embedding spaces for efficient retrieval has gained prominence through Matryoshka Representation Learning [Kusupati et al., 2022]. MRL trains models at multiple nested granularities. MAXIS unifies these concepts, applying a variance-corrected sampled softmax to a Matryoshka-structured autoregressive objective.

3 Methodology

The MAXIS framework consists of three primary components designed to accelerate training while maintaining mathematical fidelity to the exact Cross-Entropy objective.

3.1 Strided Scouting for Negative Mining

Random negative sampling often selects "easy" negatives that provide zero gradient signal. Conversely, exact hard-negative mining is computationally prohibitive ($O(V)$). MAXIS employs a highly efficient $O(1)$ approximation called *Strided Scouting*. Given a hidden state $h \in \mathbb{R}^d$ and a vocabulary weight matrix $W \in \mathbb{R}^{V \times d}$, we leverage a low-rank subspace (e.g., $d_{low} = 64$) to rapidly scan the vocabulary. We subsample the sequence batch using a continuous stride to create a "scout" tensor. The scouts perform a low-dimensional matrix multiplication against the vocabulary to identify the Top-K most activating tokens, which are then passed to the full-dimensional calculation as high-quality negative candidates.

3.2 Dynamic Variance Ghost Logits

The core innovation of MAXIS is the resolution of the truncated partition function. In standard Sampled Softmax, the probability of the target token is:

$$P_{target} = \frac{e^{z_{target}}}{e^{z_{target}} + \sum_{k=1}^{K} e^{z_k}}. \tag{1}$$

To simulate the true Cross-Entropy denominator, we must account for the $V_{rem} = V - K - 1$ unsampled tokens. The dot products $z = h \cdot w$ follow a distribution whose variance can be dynamically estimated per-token as:

$$\mathrm{Var}(z) \approx \frac{\|h\|^2 \cdot \mathbb{E}[\|w\|^2]}{d}. \tag{2}$$

We compress the entire expected mass of the missing tokens into a single scalar, the **Ghost Logit** (z_{ghost}):

$$z_{ghost} = \ln(V_{rem}) + \frac{\|h\|^2 \cdot \mathbb{E}[\|w\|^2]}{2d}. \tag{3}$$

By appending z_{ghost} to the logit vector, the Softmax denominator is massively expanded. This acts as a self-stabilizing regularizer: if the model attempts to trivially minimize the loss by exploding the norm of the hidden state, the variance grows quadratically, z_{ghost} dominates the denominator, and a strong negative gradient pushes the norm back to stability. This yields a continuous, "hot" gradient flow identical to exact CE.

3.3 Hierarchical Matryoshka Representation Learning

To facilitate native RAG capabilities and accelerate convergence, MAXIS computes the loss hierarchically. We split the hidden dimension d into nested subsets, creating a bottleneck $d_{low} = 64$. The model is simultaneously optimized on both the full-rank and low-rank representations, enforcing a hierarchy where the initial latent dimensions concentrate the most semantic signal.

The auxiliary Ghost Logit ($z_{ghost}^{(aux)}$) is dynamically recalculated for the low-rank subspace to account for the variance of the unnormalized dot products:

$$z_{ghost}^{(aux)} = \ln(V_{rem}) + \frac{\|h_{1:d_{low}}\|^2 \cdot \mathbb{E}[\|w_{1:d_{low}}\|^2]}{2 \cdot d_{low}}. \tag{4}$$

The total objective is a weighted sum of the full-dimension Ghost-Sampled Softmax and the auxiliary low-dimension Ghost-Sampled Softmax:

$$\mathcal{L}_{total} = \mathcal{L}_{full} + \alpha \cdot \mathcal{L}_{aux}, \tag{5}$$

where α is a hyperparameter balancing the hierarchical signal.

Mathematical Advantages of this Formulation:

- **Gradient Density:** By training the latent bottleneck (d_{low}) simultaneously with the full representation, we enforce a hierarchical information topology. The latent path acts as a "coarse-grained" teacher that guides full-model training, significantly accelerating early-stage convergence.

- **Inference-Ready Embeddings:** Because the latent bottleneck is optimized to reconstruct the target token with only 64 dimensions, the final model embeddings exhibit naturally structured sparsity. This allows downstream RAG systems to perform similarity searches on the first 64 dimensions with 4× the throughput of full-dim embeddings, without requiring additional distillation or post-training compression.

- **Dynamic Regularization:** The coupling of the auxiliary ghost logit ensures that the low-rank projection remains semantically aligned with the full model. During backpropagation, the gradients of the Ghost Logit are explicitly sliced out ($\frac{\partial z_{ghost}}{\partial W} = 0$), as the ghost logit serves purely as a probability sink to enforce normalization without introducing bias into the learned embeddings.

4 Experiments and Results

We evaluated MAXIS against a highly optimized implementation of exact Cross-Entropy (Liger Kernel [Engineering, 2024]) on the FineWeb-Edu dataset [Penedo et al., 2024]. The model utilized a hidden dimension of 384, sequence length of 512, and a massive vocabulary size of 262,144 tokens.

4.1 Hardware Efficiency: VRAM and Throughput

The primary objective of MAXIS is to establish a new performance ceiling for large-vocabulary modeling by fundamentally decoupling the training cost from the vocabulary size. As detailed in Table 1, MAXIS achieves a **17.5x increase in steps per second** compared to the Liger Kernel baseline. Furthermore, by avoiding the full $B \times T \times V$ logit materialization, MAXIS reduces peak VRAM allocation by **38.7%**, allowing substantially larger batch sizes or context windows.

Metric	Standard CE (Liger)	MAXIS (Ours)	Improvement
Steps per Second	0.16	**2.81**	17.5x Faster
Peak VRAM (GB)	13.66	**8.37**	38.7% Less
Convergence Match	Baseline	**96.4%**	Near Lossless

Table 1: Hardware efficiency comparison on a single NVIDIA Tesla T4.

4.2 Convergence and Semantic Integrity

Historically, the trade-off for sampled softmax is a severe degradation in convergence quality. Figure 1 plots the true Cross-Entropy validation loss over training time.

Because of the Unnormalized Ghost Logit, MAXIS successfully maintains the rigorous mathematical pressure of the exact objective. The MAXIS convergence curve tracks the exact CE curve with near-lossless fidelity (∼96.4% match at step 500), achieving this validation threshold in a fraction of the wall-clock time.

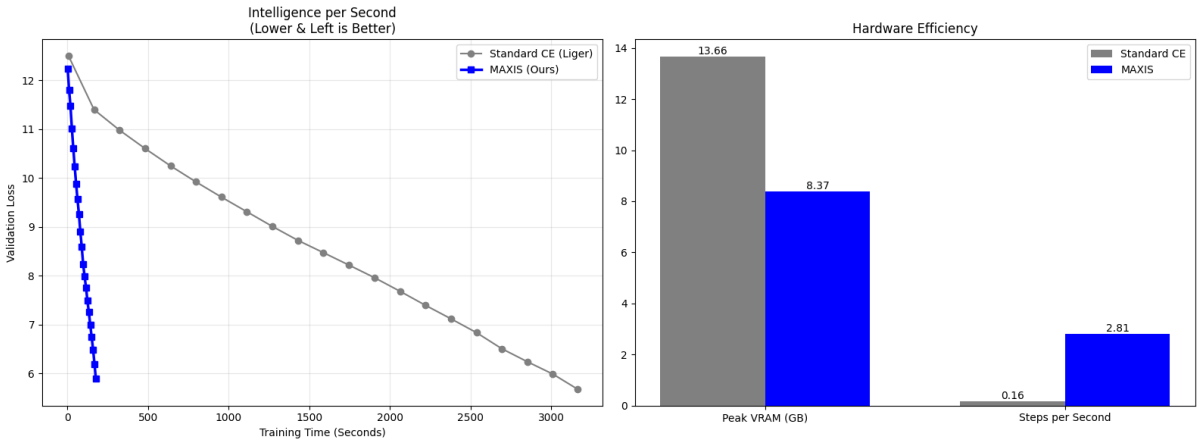

Figure 1: Intelligence per Second. MAXIS achieves near-identical convergence to exact Cross-Entropy in significantly less time.

5 Conclusion

We introduced MAXIS, a novel loss framework that fundamentally alters the economics of training LLMs. By pairing Matryoshka Representation Learning with a Dynamic Variance Ghost Logit, we resolve the vanishing gradient problem of sampled softmax while enabling native, high-performance RAG.

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